

GeneMeow GFP Design Concept and Documentation

Final submission based on DMS-positive brightness/supercharge axis and ColabFold validation

Team: GeneMeow

Summary. This final package uses six sfGFP-derived sequences from the DMS-positive/supercharge design axis. The final candidates were selected because they passed competition format checks, showed positive sfGFP-aware brightness delta scores, preserved the chromophore-forming TYG motif, and passed ColabFold core structural validation. The workflow does not claim experimental proof before official testing; it provides a reproducible computational prioritization pipeline.

1. Objective and scoring logic

The objective is to design GFP-family sequences with high initial fluorescence and high post-heat fluorescence retention. In the official assay, each sequence is synthesized, expressed in a cell-free protein synthesis system, measured for initial brightness, heat-treated, and measured again. The combined score is proportional to post-heat brightness relative to the WT control in the same batch:

$$\text{Comprehensive score} = (\text{Finitial} / \text{FinitialWT}) \times (\text{Ffinal} / \text{Finitial}) = \text{Ffinal} / \text{FinitialWT}$$

Therefore, the design strategy must protect initial brightness while increasing the chance of retaining fluorescence after heat treatment.

2. Final computational workflow

- Load official references, GFP brightness data, exclusion list and submission template.
- Use sfGFP as the parent scaffold and preserve the chromophore-forming TYG motif at positions 65-67.
- Evaluate initial brightness using an sfGFP-aware delta model derived from avGFP DMS coefficients, with corrected dataset numbering.
- Evaluate thermal-retention potential using a stability proxy based on negative surface-charge shift and mutation risk.
- Run ColabFold on WT sfGFP and all six candidates to check preservation of the GFP beta-barrel and chromophore environment.
- Submit a portfolio of conservative, balanced and thermal/supercharge candidates.

3. Brightness validation

Brightness was not treated as a directly predicted experimental Finitial value. Instead, each mutation was scored as a delta effect relative to sfGFP using avGFP DMS-derived coefficients. The key correction is that the DMS dataset numbering excludes the initial methionine, so protein position P maps to dataset position P - 1. The numbering check was perfect for the available DMS features.

All six final sequences have positive sfGFP-aware brightness delta scores and all mutation effects used in this validation were measured, not estimated. This is the main advantage of the final DMS-positive design axis.

Seq	Mutations	Brightness delta	Verdict	Measured	Estimated	Ref mismatch	Low support
1	D19E:R73L:H231Y	0.170	>= sfGFP brightness prior	3	0	0	0
2	D19E:R73L:K166E:N212D:H231Y	0.283	>= sfGFP brightness prior	5	0	0	0
3	D19E:R73L:K166E:N198D:N212D:H231Y	0.346	>= sfGFP brightness prior	6	0	0	0
4	D19E:R73L:K101E:K156E:K16	0.379	>= sfGFP brightness prior	8	0	0	0

	6E:N198D:N21 2D:H231Y						
5	D19E:R73L:H2 31Y:Y237N	0.279	>= sfGFP brightness prior	4	0	0	0
6	K101E:K156E: K166E:N198D: N212D	0.209	>= sfGFP brightness prior	5	0	0	0

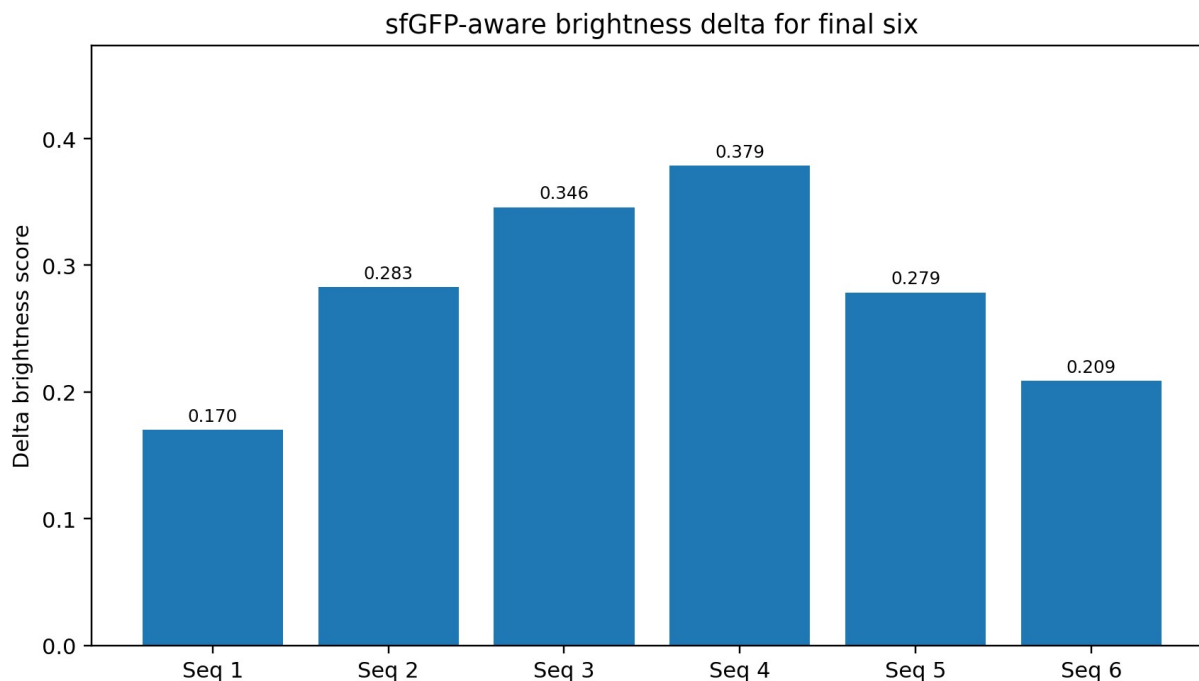


Figure 1. All six final sequences have positive sfGFP-aware brightness delta scores.

4. Stability proxy and risk control

Because no large public dataset exactly matches the official 72C heat-treatment assay, thermal retention was not predicted directly. The workflow used a stability proxy: moderate negative surface-charge shift was treated as a solubility/aggregation-resistance prior, while excessive mutational load was penalized.

The final six include conservative brightness-preserving candidates, balanced charge-shift candidates and stronger supercharge candidates. Seq 3 and Seq 6 are especially strong balanced/thermal candidates by this proxy.

Seq	Charge delta	Stability proxy	Risk penalty
1	1.10	0.316	0.240
2	4.10	1.22	0.400
3	5.10	1.51	0.480
4	9.10	1.30	1.14
5	1.10	0.316	0.420
6	8.00	1.56	0.400

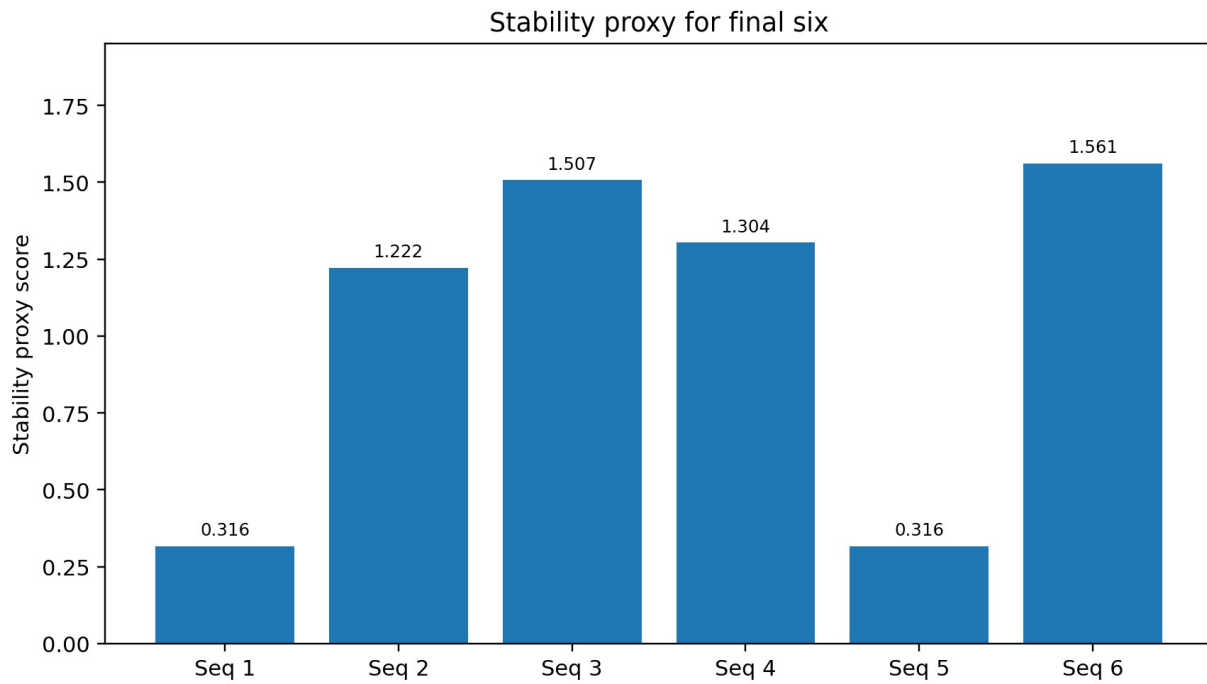


Figure 2. Stability proxy values for the final six candidates.

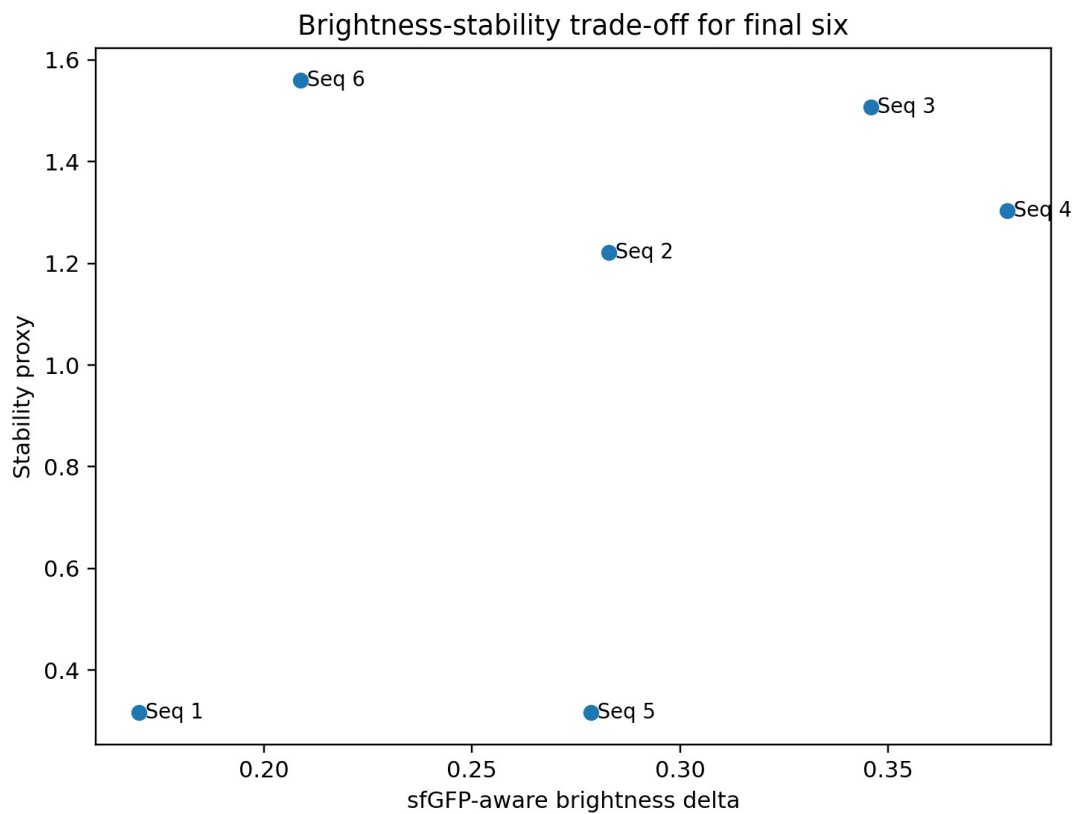


Figure 3. Brightness-stability trade-off across the final six candidates.

5. ColabFold structural validation

WT sfGFP and the six designed variants were evaluated with ColabFold AlphaFold2_batch using MMseqs2 UniRef + Environmental MSAs, five models per sequence, three recycles, no templates and no Amber relaxation. ColabFold was

used only as a structural sanity check; it does not predict fluorescence intensity, chromophore maturation or experimental heat retention.

All final candidates preserved a high-confidence GFP-like fold. Mean pLDDT values were about 96, pTM was 0.91 for all variants, chromophore-region confidence was high, and core RMSD to WT sfGFP remained below 0.13 Å. Some terminal mutations such as H231Y or Y237N have lower per-position pLDDT because the C-terminus is flexible; the core mutation-site pLDDT remains high in all variants.

Seq	Mean pLDDT	pTM	Chromo min	Core RMSD Å	Core mut-site min	Pass
1	96.21	0.910	95.38	0.125	98.12	True
2	96.17	0.910	95.75	0.040	97.75	True
3	96.23	0.910	96.00	0.042	97.12	True
4	96.33	0.910	95.31	0.123	95.88	True
5	96.33	0.910	94.88	0.128	98.12	True
6	96.22	0.910	94.75	0.115	95.75	True

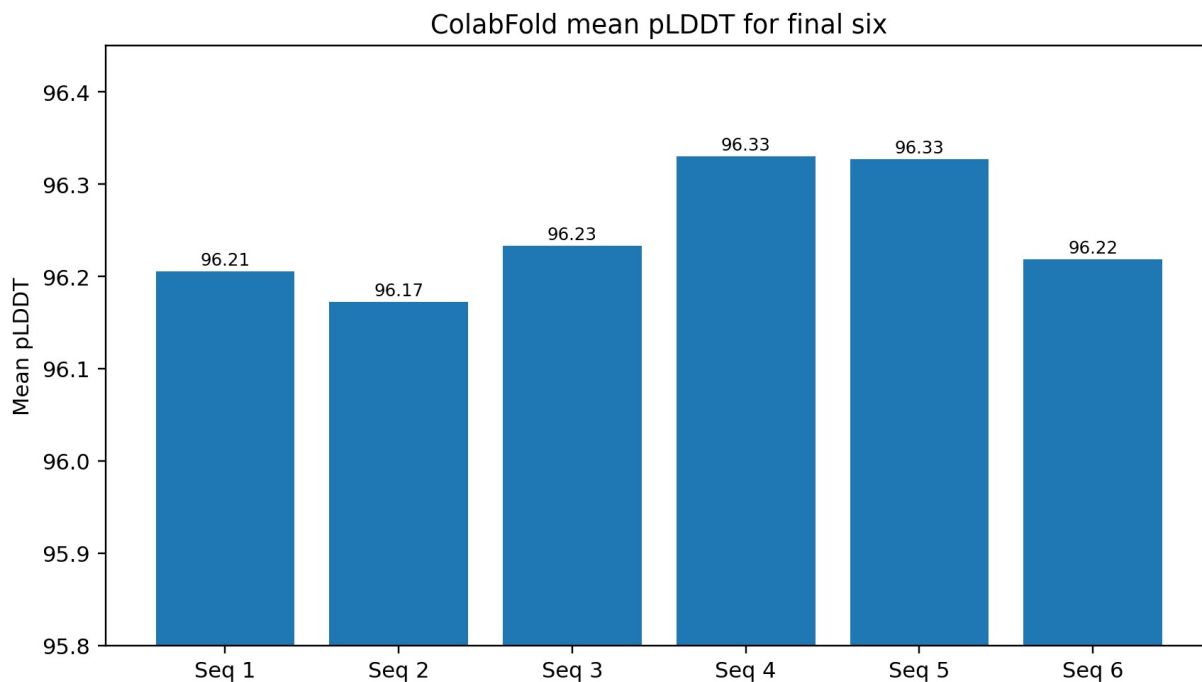


Figure 4. Mean pLDDT for the final six candidates.

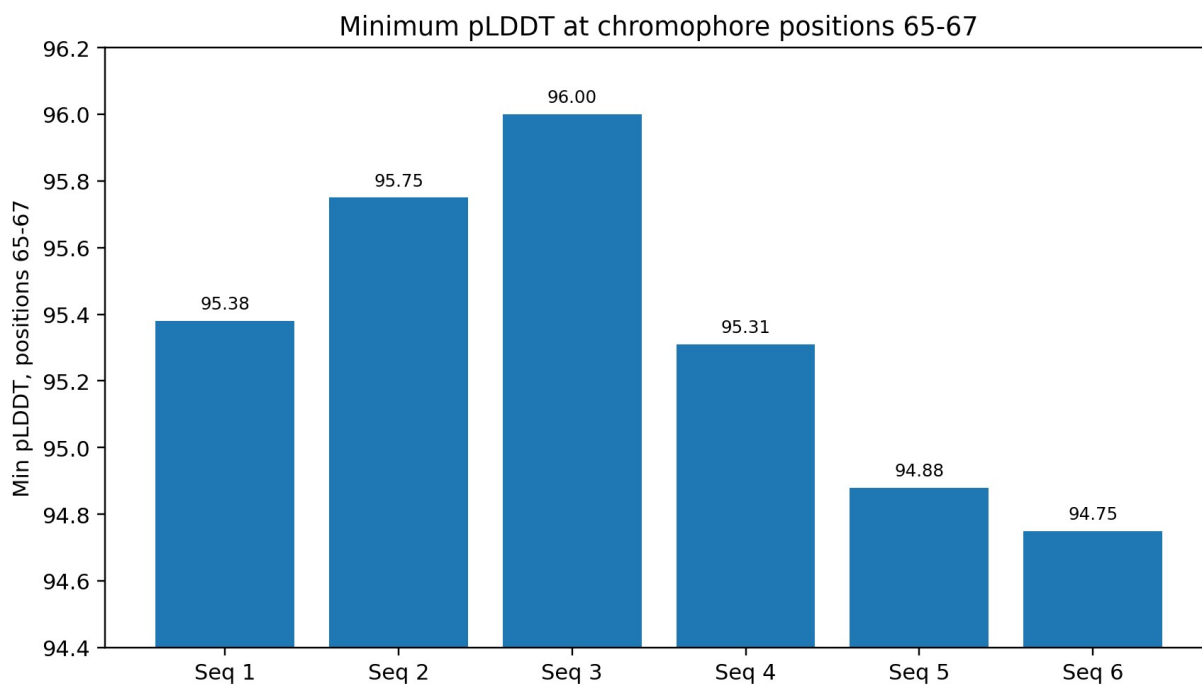


Figure 5. Minimum pLDDT in chromophore-forming positions 65-67.

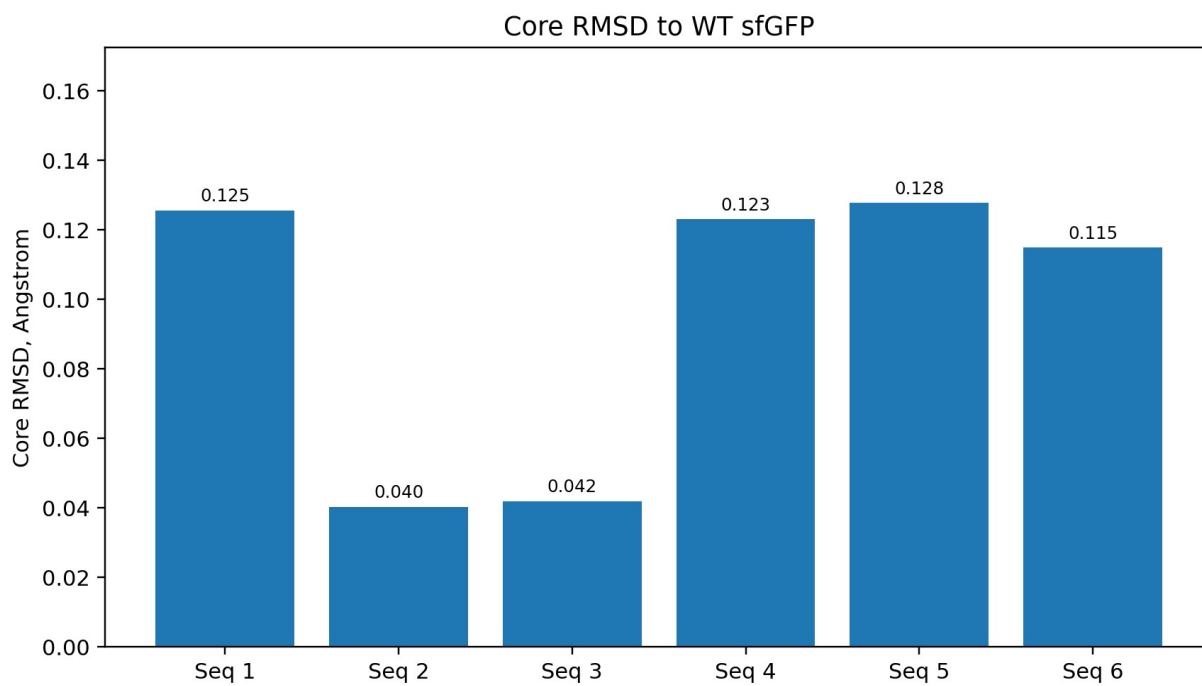
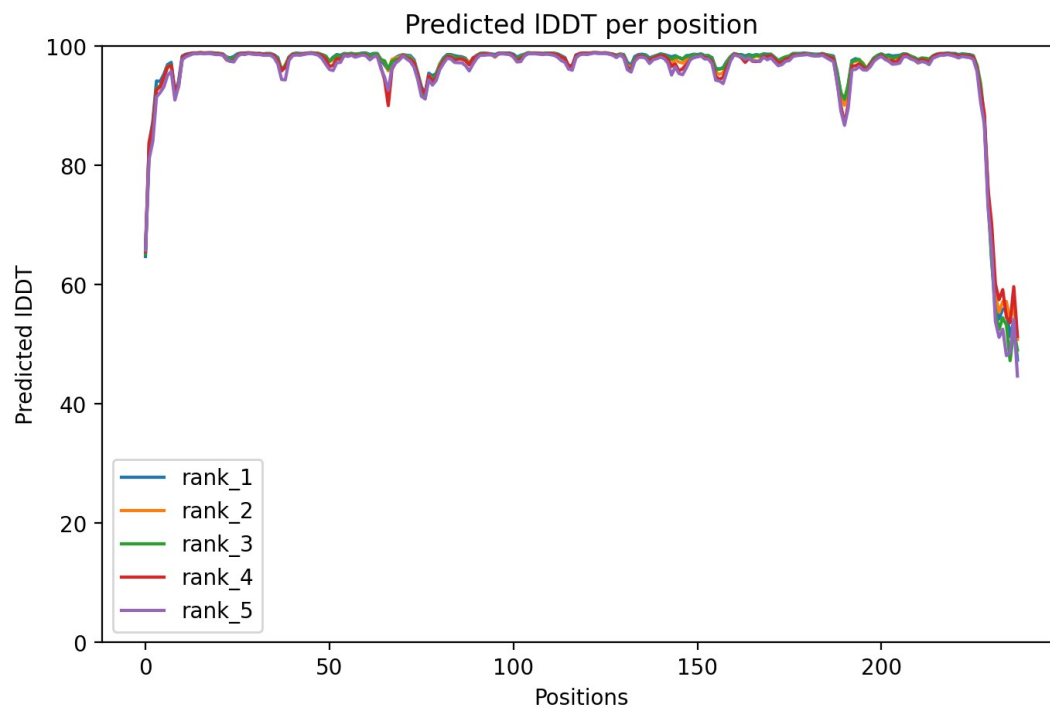


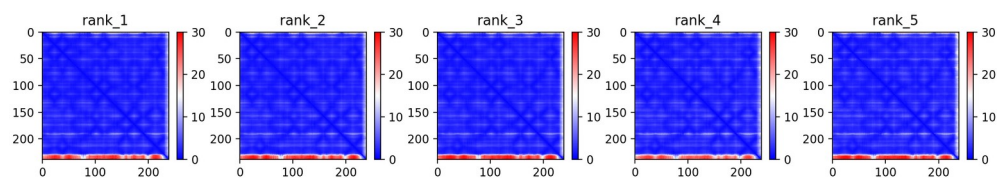
Figure 6. Core RMSD to WT sfGFP.

6. Representative ColabFold output plots

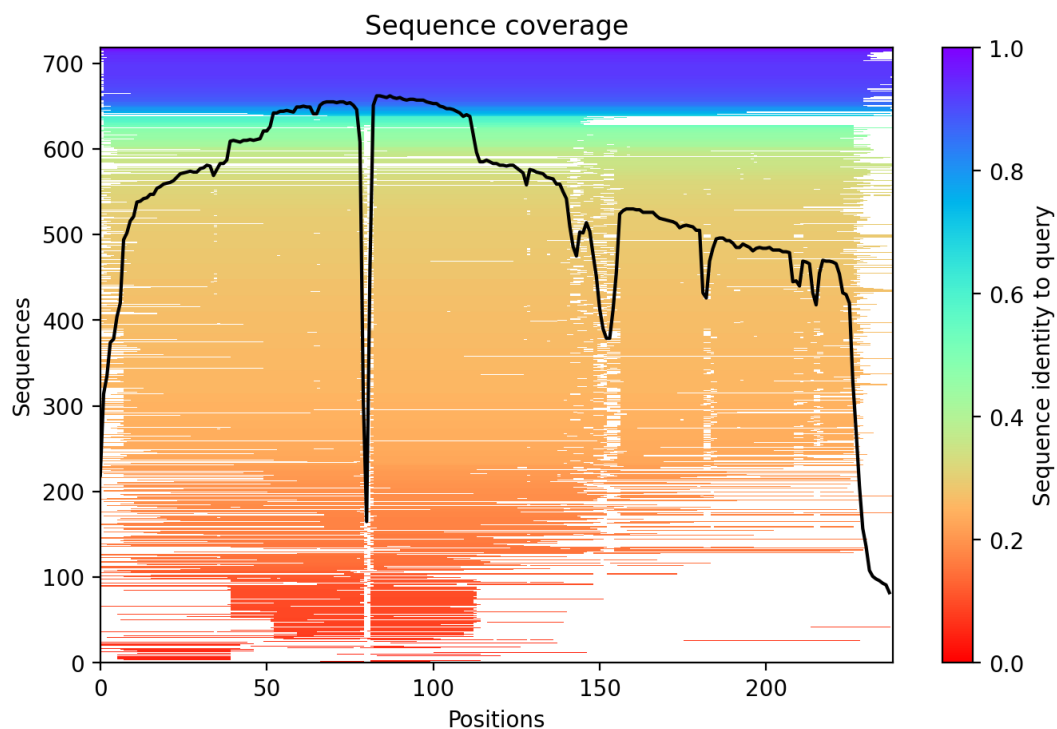
Representative raw ColabFold plots are included for the strongest balanced candidate Seq 3 and the strongest thermal/supercharge candidate Seq 6. The complete raw plots are stored in the GitHub repository.



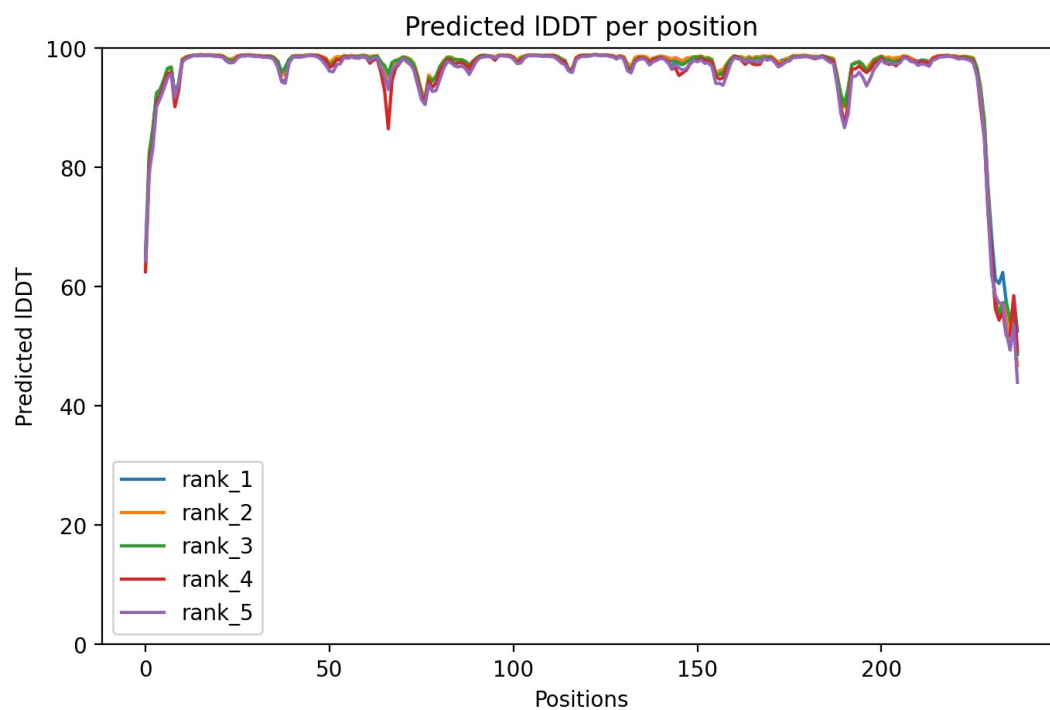
Seq 3 balanced candidate: ColabFold pLDDT plot.



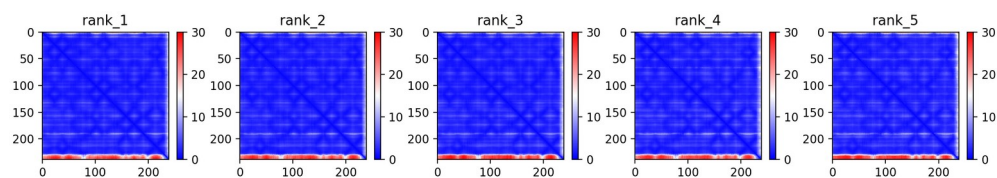
Seq 3 balanced candidate: ColabFold PAE plot.



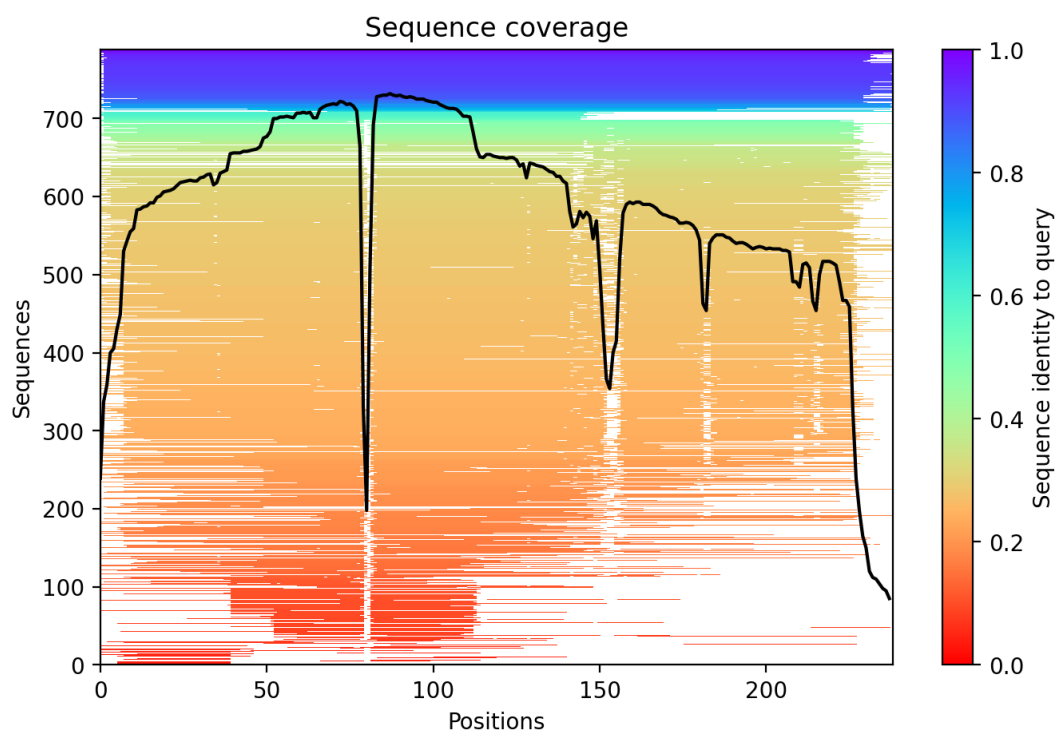
Seq 3 balanced candidate: MSA coverage plot.



Seq 6 supercharge candidate: ColabFold pLDDT plot.



Seq 6 supercharge candidate: ColabFold PAE plot.



Seq 6 supercharge candidate: MSA coverage plot.

7. Final submitted sequences

The submitted CSV contains exactly six rows with the required columns Team_Name, Seq_ID and Sequence. All sequences are 238 aa long, start with M, use only standard amino acids and preserve TYG at positions 65-67.

Seq	Role	Mutations
1	conservative surface brightness	D19E:R73L:H231Y
2	balanced core	D19E:R73L:K166E:N212D:H231Y
3	main balanced	D19E:R73L:K166E:N198D:N212D:H231Y
4	thermal high-risk/high-reward	D19E:R73L:K101E:K156E:K166E:N198D:N212D:H231Y
5	brightness backup	D19E:R73L:H231Y:Y237N
6	supercharge insurance	K101E:K156E:K166E:N198D:N212D

8. Reproducibility package

The accompanying GitHub-ready repository includes the final submission CSV, validation metrics, ColabFold plots, rank-001 PDB/JSON files, scripts for brightness scoring and ColabFold result processing, and this documentation.

9. Limitations

- Brightness delta is a DMS-derived computational prior, not a direct experimental Finitial measurement.
- The stability proxy is not a melting temperature and does not directly predict Ffinal after 72C heat treatment.
- ColabFold validates structural plausibility but does not model chromophore maturation, quantum yield, extinction coefficient, or CFPS expression yield.
- Final performance must be determined by the official experimental assay.